

# torch.nn.utils.weight\_norm#

[https://pytorch.org/docs/stable/generated/torch.nn.utils.weight\\_norm.html](https://pytorch.org/docs/stable/generated/torch.nn.utils.weight_norm.html)

## Description:

Apply weight normalization to a parameter in the given module. Weight normalization is a reparameterization that decouples the magnitude of a weight tensor from its direction. This replaces the parameter specified by name (e.g. 'weight') with two parameters: one specifying the magnitude (e.g. 'weight\_g') and one specifying the direction (e.g. 'weight\_v'). Weight normalization is implemented via a hook that recomputes the weight tensor from the magnitude and direction before every forward() call. By default, with dim=0, the norm is computed independently per output channel/plane. To compute a norm over the entire weight tensor, use dim=None. See <https://arxiv.org/abs/1602.07868>

## Function Signature

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```
torch.nn.utils.weight_norm(module, name='weight', dim=0)  
[source]#
```

Parameters

Returns

Return type

**Returns:**

T\_module

## Code Examples

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```
>>> m = weight_norm(nn.Linear(20, 40), name='weight')  
>>> m  
Linear(in_features=20, out_features=40, bias=True)  
>>> m.weight_g.size()  
torch.Size([40, 1])  
>>> m.weight_v.size()  
torch.Size([40, 20])
```

## Notes & Warnings

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### WARNING

This function is deprecated. Use `torch.nn.utils.parametrizations.weight_norm()` which uses the modern parametrization API. The new `weight_norm` is compatible with `state_dict` generated from old `weight_norm`. Migration guide: The magnitude (`weight_g`) and direction (`weight_v`) are now expressed as `parametrizations.weight.original0` and `parametrizations.weight.original1` respectively. If this is bothering you, please comment on [pytorch/pytorch#102999](https://pytorch.org/issues/102999) To remove the weight normalization reparametrization, use `torch.nn.utils.parametrize.remove_parametrizations()`. The weight is no longer recomputed once at module forward; instead, it will be recomputed on every access. To restore the old behavior, use `torch.nn.utils.parametrize.cached()` before invoking the module in question.

## Detailed Documentation

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### Documentation

Apply weight normalization to a parameter in the given module.

Weight normalization is a reparameterization that decouples the magnitude of a weight tensor from its direction. This replaces the parameter specified by name (e.g. 'weight') with two parameters: one specifying the magnitude (e.g. 'weight\_g') and one specifying the direction (e.g. 'weight\_v'). Weight normalization is implemented via a hook that

recomputes the weight tensor from the magnitude and direction before every forward() call.

By default, with dim=0, the norm is computed independently per output channel/plane. To compute a norm over the entire weight tensor, use dim=None.

See <https://arxiv.org/abs/1602.07868>

This function is deprecated. Use `torch.nn.utils.parametrizations.weight_norm()` which uses the modern parametrization API. The new `weight_norm` is compatible with `state_dict` generated from old `weight_norm`.

The magnitude (`weight_g`) and direction (`weight_v`) are now expressed as `parametrizations.weight.original0` and `parametrizations.weight.original1` respectively. If this is bothering you, please comment on [pytorch/pytorch#102999](https://pytorch.org/torch/issues/102999)

To remove the weight normalization reparametrization, use `torch.nn.utils.parametrize.remove_parametrizations()`.

The weight is no longer recomputed once at module forward; instead, it will be recomputed on every access. To restore the old behavior, use `torch.nn.utils.parametrize.cached()` before invoking the module in question.

module (Module) – containing module

name (str, optional) – name of weight parameter

dim (int, optional) – dimension over which to compute the norm

The original module with the weight norm hook

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